

Safety Training

Staff and Senior Students of Mechanical and
Mechatronic Engineering Building



General Safety Rules

Safety Procedures for Laboratory Setups



General Safety Rules

- ▶ Applies to all **students, staff** and **all other persons** present in the labs and building
- ▶ Safety committee
 - ▶ MM: Cobus Zietsman, Gillian Cortereal, Merentia Meyer, Shiyaam Valentyn
 - ▶ Various members from Industrial Engineering



General Safety Rules

- ▶ According to OHSA students are viewed as employees

Important

All students/visitors must obey instructions of lecturer/ technician or other staff member regarding safety. Not allowed to refuse according to the law.



- ▶ Working off-site, i.e. not in the engineering faculty.
 - ▶ Written permission is required from authorised site manager. Permission must state purpose, duration and scope of work and persons involved.
 - ▶ Student's supervisor will keep this on record.
 - ▶ For staff: Insurance can be obtained from:

Student Insurance

Dean Johnson
Finansiële Beplanning & Batebestuur
tel: 021 808 9195
e-pos/e-mail: deanj@sun.ac.za



- ▶ General off-campus safety guidelines
 - ▶ You must comply with the safety rules of the facility you are working at.
 - ▶ Make sure you get a safety induction before you do any practical work or enter any high risk area.
 - ▶ If you think it is unsafe, get out!
 - ▶ Travel safely to and from the workplace. Make sure you know the safest route and have safe transport.
 - ▶ Avoid travelling at night, especially in an area you are not familiar with.
 - ▶ Avoid travelling and working alone.

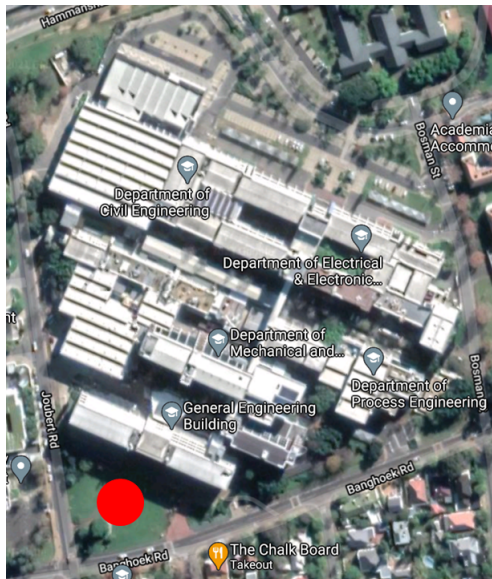


Evacuation Procedures

- ▶ When siren sounds or instructed by safety officer
 - ▶ Leave building immediately through nearest exit
- ▶ Immediately obey safety officers during exercises
- ▶ Last person leaving a room: close door and tie something (a handkerchief) to the handle
- ▶ Do not use the elevator
- ▶ Assemble on the lawn in front of the General Engineering Building
 - ▶ Keep lanes open for emergency vehicles.



Assembly Area



General Safety Rules

► **Fire/Explosion/Bomb scare**

- Call Campus Security 021 808 4666 or emergency number 021 808 2333
- Help/stay with injured, otherwise evacuate
- Do not try to extinguish fire if it endangers your health

► **Power cuts**

- Prepare for when power comes on again
- Always consider all connections to be live

Good Idea

Save Campus Security number on your cell phone



Handling of Hazardous Materials

- ▶ University policy
- ▶ Following must be disposed of in approved ways:
 - ▶ Computer equipment (Merentia Meyer)
 - ▶ Batteries (Instrumentation Store)
 - ▶ Chemicals (Cobus Zietsman)
 - ▶ Biological matter (Cobus Zietsman)



Laboratory Practicals

- ▶ Each prac must start with going through safety instructions for that prac
- ▶ Attendance register for each prac
 - each student must sign that he/she understands the safety procedures



Your Responsibility

If prac does not start with a safety briefing, please demand that it does



Laboratory Areas

- ▶ No unauthorized persons may enter a lab or workshop
 - ▶ Keep doors closed, even during office hours
 - ▶ Only access demarcated areas in the workshop
- ▶ No-one may work alone in a lab (excluding offices in labs). Also applies to after hours
 - ▶ At least one other person must be within hearing distance
- ▶ Avoid internal combustion engine lab if fire alarm sounds
 - ▶ Automatic fire extinguishers can lead to suffocation

Remember

All hours emergencies call 021 808 2333 - Always have a cell phone with you



Reporting of Incidents

- ▶ Report all safety risks to the technician/lecturer/chairperson or Campus Security after hours
 - ▶ eg. unsafe/faulty electrical wiring or cables, compressed air pipes/hoses, etc.
- ▶ Report all incidents in which a person is injured immediately to the technician/lecturer/chairperson
 - ▶ Legally required reporting procedures apply
 - ▶ Any injury that may possibly require medical treatment or have long term effect
- ▶ If fatal (or potentially fatal) injuries occur - law specifies that you must not disturb the site
- ▶ Do not hesitate to report a "near miss". It can prevent a more serious incident in future.



A collage of various styles of safety shoes. At the top left is a white leather oxford shoe with a black toe cap. To its right is a black leather safety boot with orange laces and a yellow label. Below the white shoe are three more safety shoes: a grey one, a red one, and a light blue one, all with black laces and thick black soles. A large, semi-transparent red arrow points diagonally from the top right towards the bottom left, passing over the shoes. To the right of the arrow are two brown leather safety shoes with a woven pattern. At the bottom right are two pairs of casual safety shoes: one pair is orange with white soles, and the other is dark blue with white soles. A faint watermark 'Big Cat Safety' is visible in the center of the collage.

A collage of various shoes including sandals, sneakers, and flats, with a large red 'X' overlaid, indicating that these are not recommended footwear for the workplace.

No cellphone usage



No cellphone usage when using dangerous equipment.



Fire Risks

- ▶ No earphones while working in lab or workshop (alarms may not be audible).
- ▶ Should the fire risks associated with an experiment be deemed too significant by the responsible person(s), testing may be limited to set hours (normally office hours).
- ▶ All fire risks must be identified in the safety report.
- ▶ Evacuation routes and plans must be included in all safety reports, with special emphasis on after hour work.
- ▶ A Hot Work Permit must be completed and approved before work with an open flame may start. The form is available on the Department's SUNLearn page.



Permissions Required

- ▶ Students must obtain permission before they work in any laboratory or work with any equipment
 - ▶ Obtain permission from the person in charge of that equipment or lab (given on the door of many labs; otherwise ask lecturer or lab manager)



- ▶ **If you build up a new lab setup or change an existing one**
 - ▶ You must compile safety instructions before starting to operate the setup
 - ▶ Must obtain approval of instructions from supervisor, lecturer responsible for that lab, lab engineer and lab manager (Mr Cobus Zietsman) before starting to operate the setup
 - ▶ Beware of preliminary testing in the workshop
- ▶ **If you use an existing lab setup**
 - ▶ Obtain its safety instructions or compile a new set



Safety Procedures for Laboratory Setups

See *SUNLearn* → Meganiese Ingenieurswese - Mechanical Engineering - 414 → Safety Documents

- ▶ Applies to all Mechanical and Mechatronic Projects, as well as all postgraduate students
- ▶ Applies to each new/changed laboratory setup or where the procedure for an existing set-up is not available
- ▶ Applies to any work done in our workspace, including work by people from other departments or organisations.



Safety Procedures for Laboratory Setups

- ▶ **Step 1:** Determine the potential risks the setup holds
- ▶ **Step 2:** Determine the steps that can be reasonably accomplished to minimize the risks
- ▶ **Step 3:** Design the laboratory setup with mitigating/minimizing the risks in mind
- ▶ **Step 4:** Compile safety and operating instructions with the above-mentioned in mind
- ▶ **Step 5:** Compile housekeeping instructions with the above-mentioned in mind



Safety Procedures for Laboratory Setups

- ▶ **Step 6a:** Present the design, safety instructions and housekeeping instructions to your project supervisor
 - ▶ The supervisor must approve the design and safety procedures by signing your document
- ▶ **Step 6b:** Obtain approval from the lab lecturer and/or lab engineer responsible for the laboratory
 - ▶ Signature required
- ▶ **Step 6c:** Original signed document handed over to the laboratory manager (Cobus Zietsman)
- ▶ **Step 6d:** Copy of the signed document must be displayed next to the laboratory setup



Safety Procedures for Laboratory Setups

- ▶ Step 1: Determine the potential risks the setup holds
 - ▶ The health and safety of the people who are going to use the setup
 - ▶ The health and safety of the people who are in close proximity to the setup
 - ▶ The equipment itself, the environment, surrounding buildings, etc
- ▶ Examples of risks that must be considered
 - ▶ Uncontrolled accumulation of flammable gases
 - ▶ Inadequate protection from fast rotating masses, such as shafts and fans
 - ▶ Inadequate insulation of electrical equipment, especially when working with water



Safety Procedures for Laboratory Setups

Figure: Estimating the Severity of a Risk

LIKELIHOOD (THE PROBABILITY OF AN INCIDENT WHEN THE EVENT DOES OCCUR)	
	VALUE
MIGHT AS WELL BE EXPECTED (HAPPENS OFTEN)	10
QUITE POSSIBLE	6
UNUSUAL, BUT QUITE POSSIBLE	3
ONLY REMOTELY POSSIBLE (HAS HAPPEND SOMEWHERE)	1
CONCEIVABLE, BUT VERY UNLIKELY (HASN'T HAPPENED YET)	0.5
PRACTICALLY IMPOSSIBLE (ONE IN A MILLION)	0.2
VIRTUALLY IMPOSSIBLE (APPROACHES THE IMPOSSIBLE)	0.1
EXPOSURE (THE FREQUENCY OF OCCURRENCE OF THE EVENT)	
CONTINUOUS	10
FREQUENT (DAILY)	6
OCCASIONAL (WEEKLY)	3
UNUSUAL (MONTHLY)	2
RARE (A FEW PER YEAR)	1
VERY RARE (YEARLY)	0.5
NO EXPOSURE	0.1
CONSEQUENCE	
CATASTROPHIC (MANY FATALITIES, OR DAMAGE OVER - R 10 000 000,00)	100
DISASTER (A FEW FATALITIES, OR DAMAGE OVER - R1 000 000,00)	40
VERY SERIOUS (ONE FATALITY, OR DAMAGE OVER - R 100 000,00)	15
SERIOUS (SERIOUS INJURY, PERMANENT DISABILITY, OR DAMAGE OVER- R 10 000,00)	7
IMPORTANT (TEMPORARY DISABILITY, OR DAMAGE OVER- R 1 000,00)	3
NOTICEABLE (MINOR FIRST AID, OR DAMAGE OVER - R 100,00)	1
NEGLEGIBLE	0.4
RISK SCORE	RISK CLASSIFICATION
OVER 400	VERY HIGH RISK: CONSIDER DISCONTINUING OPERATION
200- 400	HIGH RISK: IMMEDIATE CORRECTION REQUIRED
70-200	SUBSTANTIAL RISK: CORRECTION NEEDED
20-70	POSSIBLE RISK: ATTENTION INDICATED
UNDER 20	ACCEPTABLE RISK



Safety Procedures for Laboratory Setups

Risk Score = Likelihood $\times$ Exposure $\times$ Consequence
(Values from table on previous slide)



Safety Procedures for Laboratory Setups

Occupational Health and Safety Act Act 85 of 1993 as Amended

Regulations

General

- General Administrative Regulations;
- General Safety Regulations;
- Major Hazard Installation Regulations; and
- Explosives Regulations;
- Construction Regulations; and
- Regulations on Hazardous Work by Children in South Africa.

Health

- Facility Regulations
- Environmental Regulations for Workplaces;
- Lead Regulations;
- Asbestos Regulations;
- Hazardous Chemical Substances Regulations;
- Diving Regulations;
- Hazardous Biological Agents Regulations; and
- Noise Induced Hearing Loss Regulations.

Mechanical

- Lift, Escalator and Passenger Conveyor Regulations;
- General Machinery Regulations;
- Driven Machinery Regulations;
- Pressure Equipment Regulations; and
- Cert. of Competency.

Electrical

- Electrical Installation Regulations; and
- Electrical Machinery Regulations.



Safety Procedures for Laboratory Setups

- ▶ Step 2: Determine the steps that can be reasonably accomplished to minimize the risks
 - ▶ See what regulations apply (list of previous slide)
 - ▶ Safety equipment that may be needed;
 - ▶ Compile safety procedures



Safety Procedures for Laboratory Setups

- ▶ Step 3: Design the laboratory setup with mitigating / minimizing the risks in mind
- ▶ Examples of measures that can be taken
 - ▶ Protective screens on rotating machinery
 - ▶ Adequate ventilation to avoid accumulation of gases
 - ▶ Adequate insulation of electrical equipment, as well as an emergency stop
 - ▶ Interlocks to prevent access to moving parts
 - ▶ Screens over belts and chains



Safety Procedures for Laboratory Setups

- ▶ Step 4: Compile safety and operating instructions with the above-mentioned in mind
 - ▶ Use personal protective equipment
 - ▶ Maintaining good housekeeping at all times
 - ▶ Purging procedures when using flammable gasses/fuels
- ▶ Step 5: Compile housekeeping instructions with the above-mentioned in mind



Safety Procedures for Laboratory Setups

- ▶ Step 6a: Present the design, safety instructions and housekeeping instructions to your project supervisor
 - ▶ The supervisor must approve the design and safety procedures by signing your document
- ▶ Step 6b: Obtain approval from the lecturer and/or technician responsible for the laboratory
 - ▶ Signature required
- ▶ Step 6c: Original signed document handed over to the laboratory manager (Cobus Zietsman)
- ▶ Step 6d: Copy of the signed document must be displayed next to the laboratory setup



Safety Procedures for Laboratory Setups

Very Important

Laboratory work is not allowed unless these steps have been completed



Safety Procedures for Laboratory Setups

Very Important

Laboratory work may not commence before submitting a signed indemnity form (see module framework).



Conclusion

- ▶ Be wise
- ▶ Do not take risks
- ▶ Be pro-active



Onthou - Spyt kom te laat!

I am my brother's keeper

I could have saved a life that day
But I chose to look the other way
It wasn't that I didn't care
I had the time -and I was there
But I didn't want to seem a fool
or argue over a safety rule
I knew he'd done the job before
If I called it wrong, he might get sore
The chances didn't seem that bad,
I've done the same -he knew I had
So I shook my head and walked on by
He knew the risks as well as I
He took the chance, I closed an eye
And with that act -I let him die
I could have saved a life that day



I am my brother's keeper

But I chose to look the other way
Now every time I see his wife,
I'll know I should have saved his life
That guilt is something I must bear
But it isn't something YOU need to share
If you see a risk that others take
That puts their health or life at stake
The question asked, or thing you say
could help them live another day
If you see a risk and walk away,
Then I hope you never have to say
"I COULD HAVE SAVED A LIFE THAT DAY
BUT I CHOSE TO LOOK THE OTHER WAY"

